INDEX NUMBERS

An index number is a special type of average designed to measure the changes in the level of a phenomenon with respect to time, space, or some other characteristic of that type. It is used to study variations in price, changes in the production in the agricultural as well as industrial sector, increases and decreases in the imports and exports of various commodities etc. it finds application mainly in the economic sphere and enables the economists as well as the administrator to feel the pulse of the economy. Because of this index numbers are sometimes referred to as economic barometers.

Index number is aspecial average. But it is basically different from the ordinary averages. Index numbers enables us to get a representative value of a group of observations which may be used for comparison with another group even when the group consists of measurements expressed in different units or when the group is composed of different types of items.

Eg. If we want to compare the cost of living of people in a city for two periods and we have information regarding the amounts consumed of the various consumer goods like rice, wheat, oil, cloth etc each expressed in a different unit. By calculating cost of living index number we can do the comparison. That is why it is said that index number is special type of average.

INDEX NUMBERS

Construction of index numbers

Let $X_1, X_2,..... X_n$ *be the set of* variables and if we want to study the changes in the values of these variables taken as a whole between two points in time or between two places. The values of some of the variables have increased, some might have decreased and some might have remained unchanged. We require a single figure which measures the total change in the set of variables considered. The specified period oe place with reference to which we want to study the change in the values of the variables is called the the base period or base place. If X_1, $X_2,..... X_n$ are the prices of n consumer goods and if we want to study the price level of these goods in 2010 in comparison to 2000. then 2000 is called the base period and 2010 is called the current period. Let $X_{10}, X_{20},..... X_{n0}$ be the values of $X_1, X_2,..... X_n$ in the base period and $X_{1k}, X_{2k},..... X_{nk}$ be the values of $X_1, X_2,.....$ X_n in the current period. The ratio of the values of the variable in the current period to its value in the base period ie, $\frac{X_{jk}}{X_{j0}}$, j= 1,2,...n is called a relative.

INDEX NUMBERS

This ratio is independet of the units of measurement. A weighted or unweighted average of these price relatives is called an index number. If unweighted averages are used we get simple index numbers and if weighted averages are used we get weighted index numbers. Depending on the average chosen and the weights chosen we can have different index numbers.

In general P_0denote the price of a commodity in the base year and P_k denote the price in the current year. $\frac{P_k}{P_0}$ is the price relative. A weighted or unweighted average of these relative will give an index number.

INDEX NUMBERS

Simple index numbers

1. Simple Arithmetic Mean Index Number

The simple arithmetic mean of the price relative expressed as a percentage is called simple arithmetic mean index number.

Simple Arithmetic Mean Index Number = $\frac{1}{n}\sum\frac{P_k}{P_0}$ x 100

2. Simple Geometric Mean Index Number

The simple geometric mean of the price relative expressed as a percentage is called simple geometric mean index number.

Simple Geometric Mean Index Number = $\sqrt[n]{\prod\frac{P_k}{P_0}}$ x100

3. Simple Harmonic Mean Index Number

Simple harmonic mean of the price relatives expressed as percentage is called the simple harmonic mean index number.

Simple Harmonic Mean Index Number = $\frac{n}{\sum\frac{P_k}{P_0}}$ x100

INDEX NUMBERS

Similarly we can have median and mode index numbers defined as the median and mode of the relatives expressed as percentage.

4. Simple Aggregate index number

This is defined as the ratio of the sums of the prices in the current year and base year expressed as a percentage.

Simple Aggregate index number = $\frac{\sum P_k}{\sum P_0}$ x100

INDEX NUMBERS

Eg.the following table gives the prices of 6 commodities in 1970 and 1980. find the simple index numbers for 1980 taking 1970 as base year.

Commodities	A	B	C	D	E	F
Price in 1970	40	60	20	50	80	100
Price in 1980	50	60	30	70	90	110

INDEX NUMBERS

WEIGHTED INDEX NUMBERS

Weighted index numbers are obtained by taking weighted averages of the relatives. Weights may be assigned in various ways and so there are a large number of weighted index numbers.

1. Laspeyre's index number

- Let p_0 and p_k be the prices and q_0 and q_k be the quantities of the goods in the base year and curret year respectively. p_0q_0 is the money value of the goods consumed in the base year. The weighted A.M. of the price relatives taking $p_0q_0$for each good as the weight we get the Laspeyer's index number

$$\text{Laspeyer's index number} = \frac{\sum \frac{P_k}{P_0} X p_0 q_0}{\sum p_0 q_0} \text{ X100}$$

$$= \frac{\sum p_k q_0}{\sum p_0 q_0} \text{X100}$$

INDEX NUMBERS

2. Paasche's index number

The money value of the goods consumed in the current year at base year price is p_0q_k . The weighted A.M. of the price relatives with these as weights is called the Paasche's index number.

Paasche's index number = $\frac{\sum \frac{P_k}{P_0} X p_0 q_k}{\sum p_0 q_k}$ X100

$$= \frac{\sum p_k q_k}{\sum p_0 q_k} X100$$

INDEX NUMBERS

3. Fisher's ideal index number

The geometric mean of Laspeyer's and Paasche's index numbers is called Fisher's ideal index number

Fisher's ideal index number = $\sqrt{\frac{\sum p_k q_0}{\sum p_0 q_0} X \frac{\sum p_k q_k}{\sum p_0 q_k}}$X100

eg.2. Given the following data regarding the price and quantity consumed of 4 commodities in 1970 and 1980, calculate the various weighted index numbers.

commodity	Base year(1970)		Current year(1980)	
	Price p_0	Quantity q_0	Price p_k	Quantity q_k
Rice	9	120	16	150
Wheat	12	90	14	100
Sugar	8	60	12	56
oil	6	40	10	30

Comm.	p_0	q_0	p_k	q_k	$p_0 q_0$	$p_0 q_k$	$p_k q_k$	$p_k q_0$
Rice								
Wheat								
Sugar								
oil								

INDEX NUMBERS

Tests to be satisfied by a good index number

We different index numbers. The merit of an index number depends on whether it satisfies certain mathematical tests. The following are the important tests suggested.

1. The commodity reversal test
2. Unit test
3. Time reversal test
4. Circular test
5. Factor reversal test

INDEX NUMBERS

1. The commodity reversal test

An index number is said to satisfy this test if it remains unchanged even if the order in which the commodities considered is changed. All index numbers which we have considered satisfy this test.

2. Unit test

If an index number is independent of the units in which prices and quantities are expressed it is said to satisfy this test. All index numbers considered except simple aggregate index number satisfy this test.

3. Time reversal test

Let I_{0k} denote the index number calculated with the period denoted by 0 as the base and the period denoted by k as the current and $I_{k0}$the index number calculated with the periods interchanged. An index number is said to satisfy the time reversal test if I_{0k}x I_{k0}= 1 or one is the reciprocal of the other. It is assumed that multiplication by 100 is not done.

Let us examine which of the I.Ns satisfy Time reversal test.

(1) Simple A.M.I.N

$I_{0k} = \frac{1}{n} \Sigma \frac{p_k}{p_0}$, $I_{k0} = \frac{1}{n} \Sigma \frac{p_0}{p_k}$.

$I_{0k} \times I_{k0} \neq 1$

S.A.M.I.N doesn't Satisfy Time Reversal test.

(2) S.G.M.I.N

$I_{0k} = \sqrt[n]{\Pi \frac{p_k}{p_0}}$, $I_{k0} = \sqrt[n]{\Pi \frac{p_0}{p_k}}$, $I_{0k} \times I_{k0} = 1$

satisfy time reversal test.

(3) S. H. M. I N

$$I_{ok} = \frac{n}{\sum \frac{P_o}{P_k}}, \quad I_{ko} = \frac{n}{\sum \frac{P_k}{P_o}}$$

$I_{ok} \times I_{ko} \neq 1$. So doesn't satisfy.

(4) S. Ag. I. N

$$I_{ok} = \frac{\sum P_k}{\sum P_o}, \quad I_{ko} = \frac{\sum P_o}{\sum P_k}.$$

$I_{ok} \times I_{ko} = 1$. So satisfy this test.

5. Laspeyres' I.N

$$I_{0k} = \frac{\Sigma p_k q_0}{\Sigma p_0 q_0} \quad , \quad I_{k0} = \frac{\Sigma p_0 q_k}{\Sigma p_k q_k}.$$

$I_{0k} \times I_{k0} \neq 1$. dosn't satisfy.

6. Paasche's I.N

$$I_{0k} = \frac{\Sigma p_k q_k}{\Sigma p_0 q_k} \quad , \quad I_{k0} = \frac{\Sigma p_0 q_0}{\Sigma p_k q_0}$$

$I_{0k} \times I_{k0} \neq 1$. So desn't satisfy

7. Fisher's I.N.

$$I_{0k} = \sqrt{\frac{\Sigma p_k q_0}{\Sigma p_0 q_0} \times \frac{\Sigma p_k q_k}{\Sigma p_0 q_k}}$$

$$I_{k0} = \sqrt{\frac{\Sigma p_0 q_k}{\Sigma p_k q_k} \times \frac{\Sigma p_0 q_0}{\Sigma p_k q_0}}$$

$$I_{0k} \times I_{k0} = 1$$

So satisfy this test

For the problem given in eg. 2. examine which of the I.Ns satisfy Time reversal test.

Ans

L.T.N

$$I_{ok} = \frac{\Sigma P_k q_o}{\Sigma P_o q_o} = \frac{4300}{2880}$$

$$I_{ko} = \frac{\Sigma P_o q_k}{\Sigma P_k q_k} = \frac{3178}{4772}$$

$$I_{ok} \times I_{ko} = \frac{4300}{2880} \times \frac{3178}{4772} \neq 1.$$

Paaschis I.N

$$I_{ok} = \frac{\Sigma P_k q_k}{\Sigma P_0 q_k} = \frac{4772}{3178}$$

$$I_{k0} = \frac{\Sigma P_0 q_0}{\Sigma P_k q_0} = \frac{2880}{4300}$$

$$I_{0k} \times I_{k0} = \frac{4772}{3178} \times \frac{2880}{4300} \neq 1.$$

So doesnt satisfy.

(3) Fischer's I.N.

$$I_{0k} = \sqrt{\frac{\sum P_k q_0}{\sum P_0 q_0} \times \frac{\sum P_k q_k}{\sum P_0 q_k}} = \sqrt{\frac{4330}{2880} \times \frac{4772}{3178}}$$

$$I_{k0} = \sqrt{\frac{\sum P_0 q_k}{\sum P_k q_k} \times \frac{\sum P_0 q_0}{\sum P_k q_0}} = \sqrt{\frac{3178}{4772} \times \frac{2880}{4330}}$$

$$I_{0k} \times I_{k0} = 1.$$

So satisfies time reversal test

INDEX NUMBERS

4. Circular test

This is an extention of time reversal test. If an index is constructed for the year A with B as base and for the year B with C as base, the test says that the index for the year A with C as base should be the product of the first two indices. In other words if 0, 1, 2 are three years and I_{01}, I_{12}, $I_{02}$are the indices for year 1 with 0 as base , year 2 with 1 as baseand year 2 with 0 as base respectively, the circular test is said to be satisfied if $I_{01} \text{x}\, I_{12} = I_{02}$ or $I_{01} \text{x}\, I_{12} \text{x}\, I_{20}$= 1. this test becomes more important when we are not merely interested in the comparison of two years but in comparison of price changes over a period of years.

Laspeyre's, Paasche's and Fisher's index numbers do not satisfy this test. Only simple aggregate index number and simple geometric mean index number satify this test.

0, 1, 2, 3, 4, 5 - - -

$I_{01} \times I_{12} \times I_{23} \times I_{34} = I_{04}$

INDEX NUMBERS

5. Factor reversal test

This is applicable only to weighted index numbers. Let I_{pq} be the index number calculated with p as price and q as quantity and I_{qp} denoting index number obtained by interchanging p and q. The index number is said to satisfy this test if I_{pq}x I_{qp}= $\frac{\sum p_k q_k}{\sum p_0 q_0}$. That is an index number is said to satisfy this test if the product of the price index and quantity index is equal to the ratio of the total money values in the current year and base year. Let us examine which of the index numbers satisfy this test.

1. <u>L.I.N</u> $\quad I_{pq} = \frac{\Sigma P_k q_0}{\Sigma P_0 q_0}$, $\quad I_{qp} = \frac{\Sigma q_k P_0}{\Sigma q_0 P_0}$

$I_{pq} \times I_{qp} \neq \frac{\Sigma P_k q_k}{\Sigma P_0 q_0}$. So L.I.N doesn't satisfy this test

2. P.I.N $\quad I_{pq} = \dfrac{\sum p_k q_k}{\sum p_o q_k}$, $\quad I_{qp} = \dfrac{\sum q_k p_k}{\sum q_o p_k}$.

$I_{pq} \times I_{qp} \neq \dfrac{\sum p_k q_k}{\sum p_o q_o}$. So doesn't satisfy

3. F.I.N $\quad I_{pq} = \sqrt{\dfrac{\sum p_k q_o}{\sum p_o q_o} \times \dfrac{\sum p_k q_k}{\sum p_o q_k}}$

$$I_{qp} = \sqrt{\dfrac{\sum q_k p_o}{\sum q_o p_o} \times \dfrac{\sum q_k p_k}{\sum q_o p_k}}$$

F.I.N satisfy factor reversal test

$$I_{pq} \times I_{qp} = \sqrt{\dfrac{\sum p_k q_o}{\sum p_o q_o} \times \dfrac{\sum p_k q_k}{\sum p_o q_k}} \times \sqrt{\dfrac{\sum q_k p_o}{\sum q_o p_o} \times \dfrac{\sum q_k p_k}{\sum q_o p_k}} = \underline{\underline{\dfrac{\sum p_k q_k}{\sum p_o q_o}}}$$

INDEX NUMBERS

Steps in the construction of an index number

1. Defining the purpose for which the index number is constructed.

Index numbers are constructed for various purposes. The method of construction is decided after taking into consideration the purpose for which it is going to be used.clear understanding of the purpose will give an orientation regarding the individuals from whom the information is to be collected, the nature of information required, the methods to be adopted etc and so purpose should be clearly defined in the beginning itself.

2. Selection of the base year

The base year chosen should be a normal year free from wars, famines, natural calamities etc. and it should not be too distant from the current period. The values of the variable should not be too large or too small in the base year compared to other years.

INDEX NUMBERS

3. Selection of the number of items

It is possible to include all relevant items in the construction of an index number. We have to be satisfied with a set of representative items depending on the taste, habits, and customs of the people. The index maker's experience and knowledge of the data under investigation and his capacity to judge correctly will count very much in deciding the items to be included.

4. Selection of the formula

There are different formulae used for the construction of index numbers. We have to decide whether simple index number or weighted index number is appropriate. If it felt that weighing is a necessity we have to decide what the weight should be.

Bias in index numbers

Certain index numbers have tendency to overestimate the changes in the values of the variates. They are said to have upward bias. Certain other index numbers show a tendency to underestimate the changes in the values of the variates. They are said to have downward bias.

INDEX NUMBERS

Limitations of an index number

1. Since index numbers are constructed taking into account only a representative set of variables and also based on a set of values of the variates collected by sampling, index numbers are subject to sampling errors.
2. Only quantitative charactoristics can be considered in the construction of index numbers. So qualitative changes in the items are not reflected by index.
3. There are different index numbers and the choice is arbitrary. So errors may occur due to wrong choice.
4. Interested parties can misuse the index numbers and may manipulate so as to support their personal interests.
5. Errors may occur in the collection of data.

INDEX NUMBERS

COST OF LIVING INDEX NUMBER

Cost of living index numbers are index numbers designed to measure the average change in the cost of living of a specified class of people due to changes in the retail prices of a specified group of goods and services assuming that the standard of living doesnot change. This index number is also known as consumer price index number or retail price index number as we are considering the changes in the retail prices of same group of goods and services.

Construction

First step in the construction of a cost of living index number is the specification of the class of people (working class, middle class etc.) and the geographical area (city, rural, urban etc.) to be covered by the index number.

Next step is the determination of goods and services to be included in the study. The quantities of the goods and services consumed should also to determined. This can be done by conducting a family budget survey. A sampling method is adopted and from the from the units in the sample details regarding the nature and quantity of goods and services consumed in the base and current years are collected. The items of consumption may be divided into four broad categories – 1) Food 2) housing 3) clothing 4) miscellaneous. The items to be included in each category for the class of people under consideration will become from the result of the survey. The quantities of each item consumed in the two years will also to be obtained.

INDEX NUMBERS

The step is the collection of the retail prices of the various items in the base year and the current year. We assume that market conditions and consumer preferences remain unchanged from time period between base and current years. The price per each item may be collected directly from a sample of retail shops and the average taken as the current year price. The base year prices may be colleted from the records of the shops or the consumers or from data collected for a similar purpose in the base year.

The next step is the determination of the weights to be assigned to each item on the basis of the consumption pattern of the class of people under study.

There are two formulae used for the calculation of the consumer price index number.

1. Aggregate expenditure method.

$$\text{I.N.} = \frac{\sum p_k q_0}{\sum p_0 q_0} \text{X}100$$

INDEX NUMBERS

2. Family budget method

$$\text{I.N.}=\frac{\sum IV}{\sum V},$$

where V=$p_0 q_0$ and I= $\frac{p_k}{p_0}$ x 100

Uses of consumer price index number

1. It is used for wage contracts
2. It is used for measuring the changing purchasing power of money
3. At govt level it is used for wage policy, taxation, rent controll etc.

INDEX NUMBERS

Eg. 1. Construct cost of living I.N. from the following

Group	A	B	C	D
Index I	300	250	200	125
weight V	4	3	2	1

Total

IV

$$I.N = \frac{\Sigma IV}{\Sigma V}$$

INDEX NUMBERS

Eg. 2. sum of the weighted indices of price relatives for 1998 and 2001 are 11815 and 10945 respectively. The sum of the weights is 85. find the weighted I.N. for 1999 and 2002.

For the year 1998 $\quad \Sigma IV = 11815$

$$\Sigma V = 85$$

1999 $\quad I.N = \dfrac{\Sigma IV}{\Sigma V} = \dfrac{11815}{85} =$

2002, $\quad I.N = \dfrac{\Sigma IV}{\Sigma V} = \dfrac{10945}{85} =$

INDEX NUMBERS

Eg. 3. an enquiry into the budgets of middle class families in a city gave the following information. What changes in the cost of living of 2000 compared with 1995 are seen?

Expense on	food	rent	cloth	fuel	others	Total
	40%	10%	15%	15%	20%	100
Price(1995) P_0	50	25	70	20	35	
Price(2000) P_1	45	30	60	25	45	
$I = \frac{P_1}{P_0} \times 100$	90	120	85.7	125	128.6	ΣIV
IV	90×40	120×10	85.7×15	125×15	128.6×20	10532.5

$$I.N = \frac{\Sigma IV}{\Sigma V} = \frac{10532.5}{100} = 105.325$$

INDEX NUMBERS- Problems

- Find Laspeyer's, Paasche's and Fisher's index numbers from the data given below

Commodities	Base period		Current period	
	Price	Quantity	Price	Quantity
A	5.0	18	8.5	18
B	2.5	7	6.0	7
C	12.5	3	18.5	2
D	7.6	2	12.1	3
E	3.5	3	7.8	3

- Find Fisher's index number from the following data and also show that Fisher's index number satisfies time reversal test and factor reversal test using the given data

Commodities	Base period		Current period	
	Price	Quantity	Price	Quantity
A	6	50	10	56
B	2	100	2	120
C	4	60	6	60
D	10	30	12	24
E	8	40	12	36